Middle East Technical University
EE501 Linear System Theory I
Fall 2026-2027

Instructor: Elif Vural

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Prerequisite: Due to METU regulations, no official prerequisite exists but a good understanding of basic linear algebra (MATH260) will be assumed.

Grading: Midterm Exam 45%, Final Exam 50%, Quizzes 5%

Midterm Date: To be fixed later (around Weeks 8-10)

Make-Up Exam: A single make-up exam will be organized for students with an official excuse. The make-up exam will be held at the end of the term and will cover all subject matter like the final exam.

Study Aids: Problem sets are available on ODTÜClass for self-study.

Lecture Notes: The lectures will be conducted in the traditional format, with the material written on the board. As a supplementary resource, handwritten lecture notes from a previous year is available on ODTÜClass. These notes may contain typos or other writing errors, and should therefore not be regarded as an official version of the course material.

References:

- S. Axler, Linear Algebra Done Right, Springer, 1997.
- C. A. Desoer, Notes for a Second Course on Linear Systems, Van Nostrand Reinhold, 1970.
- C. D. Meyer, Matrix Analysis and Applied Linear Algebra, SIAM, 2000.
- G. Strang, Linear Algebra and its Applications, 4th Ed. Brooks/Cole, 2006.

Tentative Course Schedule:

Week	Subject
1	Algebraic structures defined on sets, fields and vector spaces
2	Subspaces, linear independence, basis
3	Normed vector spaces, sequences, series and convergence in normed spaces, matrix norms
4	Inner product spaces, orthogonality, Gram-Schmidt orthonormalization
5	Linear transformations, null space, range space, matrix representations
6	Direct sum decompositions, projection theorem, orthogonal projections
7	Systems of linear equations of the form Ax=b
8	Spectral analysis of linear transformations, eigenvalues and eigenvectors
9	Invariance under operators, block diagonal forms
10	Cayley-Hamilton theorem, characteristic and minimal polynomials
11	Jordan canonical forms
12	Jordan canonical forms
13	Hermitian matrices, positive definite matrices, positive semi-definite matrices
14	Functions of a matrix